THE IMAGE OF A CLOSED NOWHERE DENSE SUBSET UNDER A QUASI-FINITE MORPHISM IS NOWHERE DENSE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-image-nowhere-dense-quasi-finite is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 15253-15258 and proof span 15260-15284 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement contains no hypotheses on the base beyond the morphism being quasi-finite; the author's formulation is preserved.

Editorial note. The proof contains an xymatrix triangle displaying the factorization; it is copied verbatim and the retained chapter preamble loads Xy-pic.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The same statement for finite morphisms, established earlier in the same chapter (and by a previous card of this lane). Cited at proof lines 15260-15261 and 15283-15284. Gives the conclusion in the finite case, to which the proof reduces.
- The affine structure theorem for quasi-finite morphisms, factoring the morphism as an open immersion followed by a finite morphism, cited from the same chapter. Cited at proof lines 15270-15274. Produces the affine intermediate scheme Z over which the morphism is finite.
- Nowhere density is local on the target, a finite union of closed nowhere dense subsets is nowhere dense, and the closure of the image of a closed nowhere dense subset under an open immersion is nowhere dense. Cited at proof lines 15262-15269 and 15276-15278. Allows the reduction to an affine base, the reduction to one member of the affine cover, and the passage from the subset to its closure in the intermediate scheme.

Source reference. Original lemma 03HX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4583–4583. Same-chapter reference used in the retained author text at source line(s) 15261, 15283. It is supplied only as a navigation

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

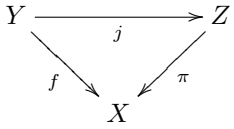
Source reference. Original lemma 03GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4541–4541. Same-chapter reference used in the retained author text at source line(s) 15270, 15522. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 03J0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4622–4622. Same-chapter reference used in the retained author text at source line(s) 15279. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 03HO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4574–4574. Same-chapter reference used in the retained author text at source line(s) 12547, 15267. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : Y \to X$ be a quasi-finite morphism of schemes. Let $T \subset Y$ be a closed nowhere dense subset of Y. Then $f(T) \subset X$ is a nowhere dense subset of X.*

Proof. As in the proof of Lemma 03HX this reduces immediately to the case where the base X is affine. In this case $Y = \bigcup_{i=1,\ldots,n} Y_i$ is a finite union of affine opens (as f is quasi-compact). Since each $T \cap Y_i$ is nowhere dense, and since a finite union of nowhere dense sets is nowhere dense (see Topology, Lemma 03HO), it suffices to prove that the image $f(T \cap Y_i)$ is nowhere dense in X. This reduces us to the case where both X and Y are affine. At this point we apply Lemma 03GU above to get a diagram



with Z affine, π finite and j an open immersion. Set $\overline{T} = \overline{j(T)} \subset Z$. By Topology, Lemma 03J0 we see $\overline{T}$ is nowhere dense in Z. Since $f(T) \subset \pi(\overline{T})$ the lemma follows from the corresponding result in the finite case, see Lemma 03HX. □